Quadratic functions and technology

1

a
- i $(8, 7)$
- ii $(0, 263)$

b
- i $(-6, -9)$
- ii $(0, 135)$

c
- i $(-6, 4)$
- ii $(0, -140)$

d
- i $(2, -1)$
- ii $(0, -13)$

e
- i $\left(-\frac{1}{2}, \frac{33}{4}\right)$
- ii $(0, 9)$

f
- i $\left(-\frac{3}{8}, -\frac{25}{16}\right)$
- ii $(0, -1)$

2

a
- i $(0, 16)$
- ii No
- iii Increasing

b
- i $(0, -12)$
- ii No
- iii Decreasing

c
- i $(0, 125)$
- ii Yes, $5, -5$
- iii Decreasing

d
- i $(0, -64)$
- ii Yes, $4, -4$
- iii Increasing

3

a No

b Yes

c Yes

d No

4 $c = 25$

5

a
- i $-\frac{5}{2}$
- ii $\frac{25}{4}$
- iii $\left(-\frac{5}{2}, 0\right)$

b
- i $1, 5$
- ii -5
- iii $(3, 4)$

c
- i $1, 7$
- ii -7
- iii $(4, 9)$

6

a
- i $(-1, 3)$
- ii $(1, 3)$

b 2 units

7 Maximum value

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

a Minimum **b** $(0, -4)$

9 $y = -\frac{81}{4}$

10

a -3.1 **b** -16.61

11

a
i $0, 6$ **ii** 0 **iii** 3 **iv** $(3, 9)$

b
i $0, -2$ **ii** 0 **iii** -1 **iv** $(-1, -1)$

c
i $0, -4$ **ii** 0 **iii** -2 **iv** $(-2, 4)$

d
i $-1, 3$ **ii** -3 **iii** 1 **iv** $(1, -4)$

e
i $-3, 1$ **ii** -3 **iii** -1 **iv** -4

f
i $2, 6$ **ii** -24 **iii** 4 **iv** 8

12

a 2.56 **b** 4.45

c
i Yes **ii** No **iii** No

d $5.12, 0.00$

13

a 2.75 **b** -9.10

c
i No **ii** Yes **iii** No

d $-0.42, 5.91$

14

a 1.55 **b** 2.95

c
i No **ii** Yes **iii** No **iv** No

d $3.58, -0.47$

15

a -1.38 **b** 0.84

c
i No **ii** No **iii** No **iv** Yes

d $-0.69, -2.08$

16 Move the graph downwards by 5 units.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

17 Move the graph to the right by 3 units.

18 Narrows the graph

19

a $y = x^2 + 2$ **b** $y = -x^2$ **c** $y = -x^2 - 2$ **d** $y = (x + 2)^2 + 2$

20 $y = (x - 2)^2 + 3$

21

a $x = 0, x = \frac{8}{7}$ **b** $x = -\frac{1}{2}, x = 3$ **c** $x = \frac{3}{2}, x = -1$ **d** $x = -\frac{1}{4}, x = 5$

e $x = -1$ **f** $x = -4, x = -3$

22

a 1 **b** 2 **c** 0

23 $2.000, 1.667$

24 $1.584, -0.859$

Additional questions

25

a $(0, 0)$ **b** The initial distance fallen by the object

26

a $0, 30$

b 15

c The largest rectangular area that can be fenced off with 60 m of wire.

27 ± 14 m/s

28

a 176.4 m **b** 6 s **c** 12 s

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

29

a $900\ \text{m}$

b Initially, the object has not fallen any distance.

c $13.55\ \text{s}$

d After $13.55\ \text{s}$ the object is on the ground.

30

a $4\ \text{s}$ **b** $10\ \text{s}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647